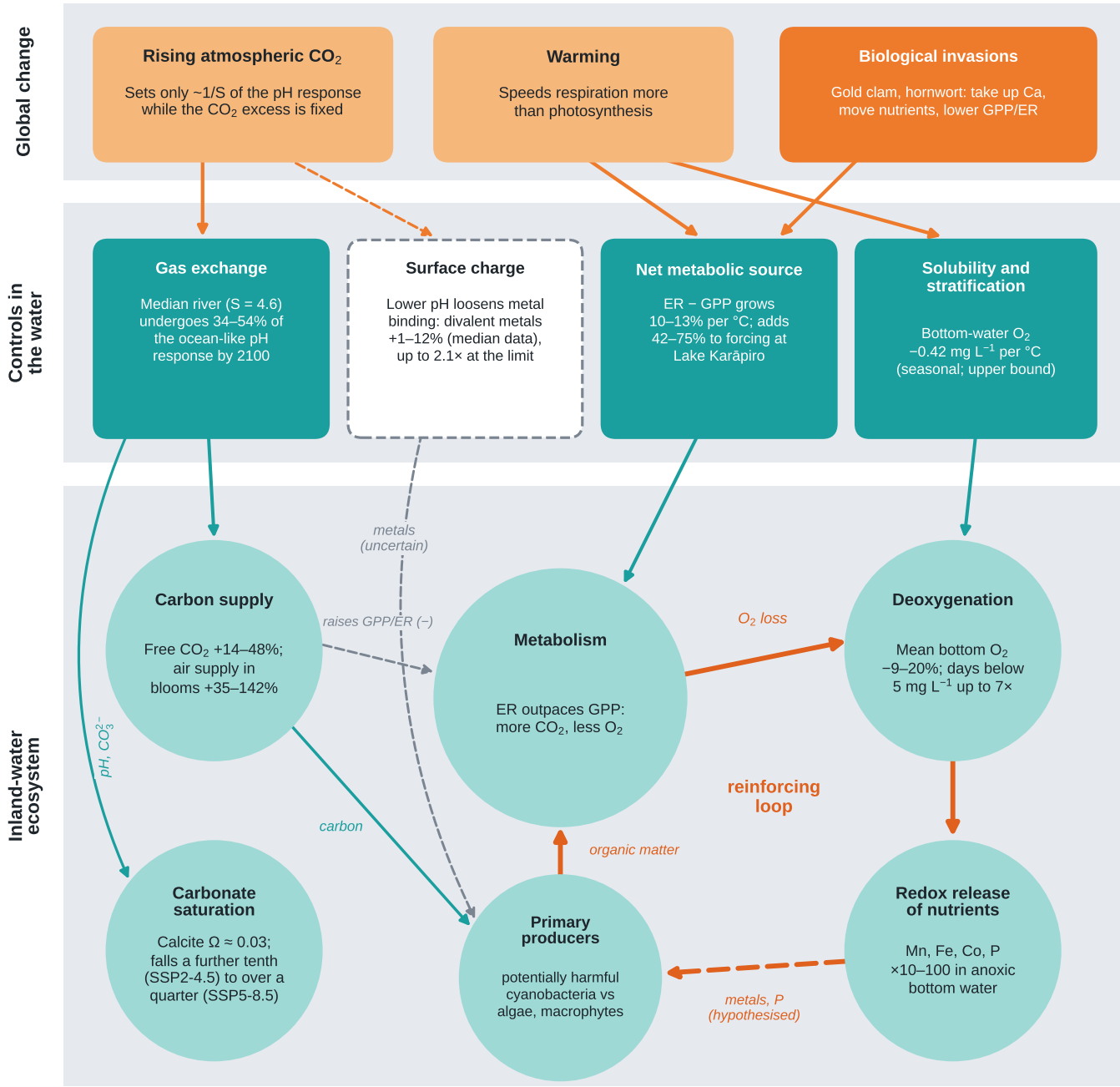

Global change
Rising atmospheric CO_2
Sets only ~1/S of the pH response while the CO_2 excess is fixed
Warming
Speeds respiration more than photosynthesis
Biological invasions
Gold clam, hornwort: take up Ca, move nutrients, lower GPP/ER
Controls in the water
Gas exchange
Median river (S = 4.6) undergoes 34–54% of the ocean-like pH response by 2100
Surface charge
Lower pH loosens metal binding: divalent metals +1–12% (median data), up to 2.1× at the limit
Net metabolic source
ER − GPP grows 10–13% per °C; adds 42–75% to forcing at Lake Karāpiro
Solubility and stratification
Bottom-water O_2 −0.42 mg L^{-1} per °C (seasonal; upper bound)
Inland-water ecosystem
metals (uncertain)
Carbon supply
Free CO_2 +14–48%; air supply in blooms +35–142%
raises GPP/ER (−)
Metabolism
ER outpaces GPP: more CO_2, less O_2
O_2 loss
Deoxygenation
Mean bottom O_2 −9–20%; days below 5 mg L^{-1} up to 7×
pH, CO_3^{2-}
carbon
reinforcing loop
organic matter
Carbonate saturation
Calcite Ω ≈ 0.03; falls a further tenth (SSP2-4.5) to over a quarter (SSP5-8.5)
Primary producers
potentially harmful cyanobacteria vs algae, macrophytes
Redox release of nutrients
Mn, Fe, Co, P ×10–100 in anoxic bottom water
metals, P (hypothesised)